Science practice booklet: Grade 5 expectations

Grade 5 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 5 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Living things

Standards practised: 5-LS1-1, 5-LS2-1

A worked example

In a mangrove forest, crabs eat fallen mangrove leaves and herons eat the crabs. Write the food chain, show which way the energy moves, and say where the energy first comes from.

1. Start with the producer: the mangrove tree makes its own food using sunlight.
2. The crab eats the leaves, and the heron eats the crab.
3. Write the chain with arrows pointing from the food to the one that eats it: mangrove leaves to crab to heron.
4. The energy first comes from the Sun, because the mangrove uses sunlight to make food.

Mangrove leaves to crab to heron; the energy first comes from the Sun.

Questions

1. Which two things from the environment do plants use to make their food?

2. Where does a plant get the energy it needs to make food?

3. A date palm seedling grows into a tall tree, but the sand around its roots hardly loses any weight. Where does most of the tree's new material come from?

4. What is a producer? Give one example from the UAE.

5. What is a consumer? Is a camel a consumer? Explain.

6. In the desert, grass is eaten by a hare and the hare is eaten by a fox. Write this food chain with arrows.

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7. What do the arrows in a food chain show?

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8. What is the difference between a food chain and a food web?

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9. What do decomposers, such as fungi and bacteria, do in an ecosystem?

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10. In a mangrove forest the number of crabs falls a lot. Name two things that might happen.

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11. Explain why almost all the energy in a food web can be traced back to the Sun.

Section 2: Earth and space

Standards practised: 5-ESS1-1, 5-ESS1-2, 5-ESS2-1, 5-ESS2-2, 5-ESS3-1

A worked example

Salem measures the shadow of a tall pole at three times of day. At 8 am it is 90 cm, at 12 noon it is 20 cm and at 4 pm it is 85 cm. When is the Sun highest in the sky, and why does the shadow change?

1. The shortest shadow is the one at 12 noon (20 cm).
2. A short shadow means the Sun is high in the sky.
3. The Earth spins, so the Sun seems to move across the sky during the day.
4. When the Sun is low (morning and afternoon) the shadow is long, and when it is high the shadow is short.

The Sun is highest at 12 noon, because the Earth spins and the Sun seems to move across the sky.

Questions

1. Why does the Sun look so much bigger and brighter than the other stars?

2. What causes day and night?

3. A flagpole's shadow is long at 7 am and short at noon. Where is the Sun in the sky at each time?

4. In the morning your shadow falls to the west. In which direction is the Sun?

5. Why do we see different stars in the night sky in winter and in summer?

6. In summer in the UAE the daylight lasts longer than in winter. What makes the length of daylight change through the year?

7. Name the four Earth systems that interact with each other.

8. Rain falls in the Hajar mountains and fills a wadi. Name two Earth systems that are interacting.

9. About 97 out of every 100 parts of Earth's water are salty ocean water. Which kind of water is most of the rest, and where is much of it kept?

10. Why does the UAE make fresh water from sea water by desalination?

11. Give two ways a family can save water at home.

12. Explain how protecting mangrove forests helps both nature and people.

Section 3: Matter, forces and energy

Standards practised: 5-PS1-1, 5-PS1-2, 5-PS1-3, 5-PS1-4, 5-PS2-1, 5-PS3-1

A worked example

Maryam stirs 30 g of salt into 200 g of water until the salt seems to disappear. What is the mass of the salty water? How could she get the salt back?

1. The salt is still there. It has broken into particles too small to see and spread through the water.
2. Matter is not lost when it dissolves, so the masses add up: 200 g + 30 g = 230 g.
3. She can heat the salty water gently so the water evaporates; the salt stays in the pot.

230 g; she can evaporate the water and the salt is left.

Questions

1. Matter is made of particles that are too small to see. Use this idea to explain how you can smell karak from across a room.

2. A balloon is full of air. Air is invisible, so how can we show that it is matter?

3. Hamdan dissolves 50 g of sugar in 250 g of water. What is the total mass of the sugar water?

4. An ice cube has a mass of 40 g. It melts completely. What is the mass of the water?

5. Salt and sand are mixed together. Describe one way to separate them.

6. Desalination plants make fresh water from sea water. Is sea water a mixture? How can the water be separated from the salt?

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7. Why would a cooking pot be made of metal and its handle of wood or plastic?

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8. Baking soda is added to vinegar and bubbles of gas form. Is this a physical change or a chemical change? How can you tell?

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9. Which is a chemical change: tearing paper, melting wax, or burning wood? Explain.

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10. In which direction does gravity pull objects on Earth? What happens to a ball you drop?

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11. A child in the UAE and a child on the opposite side of the world both drop a stone. Both stones fall down. What does this tell us about the direction of gravity?

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12. The energy in the dates Layla eats first came from the Sun. Explain how.

Answer guide for parents

Section 1: Living things

1. Water and carbon dioxide (a gas in the air), with energy from sunlight.
2. From the Sun (sunlight).
3. From water and from carbon dioxide in the air, not from the soil.
4. A living thing that makes its own food, such as a mangrove tree, a ghaf tree or a date palm.
5. A consumer cannot make its own food and eats other living things. A camel is a consumer because it eats plants.
6. Grass to hare to fox (the arrows point from the food to the animal that eats it).
7. The direction in which energy and matter move, from the food to the one that eats it.
8. A food chain shows one path of energy. A food web joins many chains together, because most animals eat more than one kind of food.
9. They break down dead plants and animals and return nutrients to the soil, so plants can use them again.
10. Herons and other animals that eat crabs may have less food; leaves may pile up because fewer are eaten. (Any sensible answer.)
11. Plants use sunlight to make food, and animals get their energy by eating plants or by eating animals that ate plants.

Section 2: Earth and space

1. It is much closer to us. The other stars are very far away.
2. The Earth spins on its axis once about every 24 hours. The side facing the Sun has day and the side facing away has night.
3. Low in the sky at 7 am and high in the sky at noon.
4. In the east (a shadow falls on the opposite side from the Sun).
5. The Earth travels around the Sun during the year, so at night we face a different part of the sky in each season.
6. The Earth is tilted as it travels around the Sun, so in summer our part of the Earth leans towards the Sun and gets more hours of daylight.
7. The geosphere (rock and land), hydrosphere (water), atmosphere (air) and biosphere (living things).
8. Any two of: the atmosphere (the rain), the geosphere (the mountains and the wadi bed), the hydrosphere (the water that flows and fills the wadi).
9. Fresh water; much of it is frozen in glaciers and ice caps, and the rest is in the ground, rivers and lakes.
10. There is little rain and there are no big rivers, but there is a lot of sea water.
11. For example: turn off the tap while brushing teeth, take shorter showers, fix leaking taps, water plants in the cool evening. (Any two sensible ways.)
12. Mangroves shelter fish, crabs and birds, and their roots help protect the shore from waves and erosion, so nature and coastal communities both benefit.

Section 3: Matter, forces and energy

1. Tiny particles of the scent travel through the air (a gas) from the cup to your nose, even though we cannot see them.
2. Air takes up space (it inflates the balloon) and it has mass, so it is matter.
3. 300 g (50 g + 250 g; nothing is lost when sugar dissolves)
4. 40 g (melting changes the state, not the amount of matter)
5. Add water to dissolve the salt, filter out the sand, then evaporate the water to get the salt back.
6. Yes, it is a mixture of water and salt. Heating makes the water evaporate; the vapour is cooled and condenses into fresh water, and the salt stays in the pot.
7. Metal conducts heat well, so the food heats up. Wood and plastic do not conduct heat well, so the handle stays cooler to hold.
8. A chemical change: a new substance (a gas) is made, shown by the bubbles.
9. Burning wood. It makes new substances (ash, smoke and gases) that cannot be turned back into wood.
10. Down, towards the centre of the Earth. The dropped ball falls to the ground.
11. Gravity pulls towards the centre of the Earth, so 'down' is a different direction in different places.
12. The date palm used sunlight to make food. Layla's body uses the energy in the dates for moving, growing and keeping warm.